House Price Prediction Using Machine Learning with Location-Based Feature Engineering

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***Abstract*—House-price estimation is challenging because market values depend not only on dwelling attributes but also on neighbourhood context, spatial accessibility, local market levels, and the geographic coverage of the training data. This paper studies location-based feature engineering for residential property-price prediction across seven Indian cities. Four public real-estate data sources are harmonised into a common schema; after cleaning, 55,912 records remain. The proposed pipeline combines property attributes, locality geocoding, coordinate-derived features, spatial clusters, hierarchical smoothed location-price encodings, and a distance-weighted spatial-neighbour signal. Five regressors—Ridge Regression, Random Forest, XGBoost, LightGBM, and CatBoost—are compared. On the notebook's random 80/20 hold-out split, the full feature set reduces median absolute percentage error (MdAPE) from 27.15% to 21.77% for LightGBM, an absolute reduction of 5.38 percentage points. A feature ablation and three geographic evaluation settings are used to distinguish ordinary random-split performance from generalisation to unseen localities and cities. Grouped SHAP attribution is used to describe model behaviour, while a few-shot experiment measures how performance changes as labelled listings become available in a target city. Results indicate that location-derived signals are useful, but geographic transfer remains difficult and should be assessed separately from random-split accuracy.**

***Index Terms*—house-price prediction, location-based feature engineering, geospatial data, target encoding, gradient boosting, spatial validation, SHAP, Indian real estate**

I. Introduction

Residential property valuation is a heterogeneous prediction problem. Properties with similar floor area and room counts may command different prices because of location, access to services, neighbourhood demand, and city-level market conditions. Hedonic price theory provides a foundation for modelling prices as a function of property characteristics and implicit attributes [1]. In machine learning, tree ensembles can capture non-linear interactions among such variables, but their performance depends on how location is represented and how evaluation data are split.

A common evaluation practice randomly divides listings into training and test sets. For location-rich data, this may place listings from the same locality in both partitions, allowing the model to exploit neighbourhood patterns that will not be available for a genuinely unseen locality. Spatial dependence is a well-established concern in predictive modelling, and validation design should reflect the intended deployment setting [2], [4]. Accordingly, this work evaluates location features under random, locality-held-out, and leave-one-city-out protocols.

A. Problem statement

Given a property listing with dwelling attributes and a locality or geographic context, estimate its sale/listing price while reducing error on both familiar and unfamiliar areas. The practical challenge is to construct location features that are informative without leaking test prices, and to report performance under evaluation protocols that reflect spatial generalisation.

B. Objectives

The study has five objectives:

1) Harmonise heterogeneous real-estate datasets into a consistent representation of price, area, rooms, location, and source.
2) Engineer transferable geographic descriptors from locality coordinates and construct a hierarchical location-price encoding with shrinkage.
3) Compare five regression algorithms using property-only and location-aware feature sets.
4) Evaluate feature contributions using ablation and geographic hold-out protocols, with locality-cluster bootstrap uncertainty estimates where available.
5) Interpret the fitted tree model using grouped SHAP attribution and assess cold-start behaviour as a new city receives labelled examples.

C. Contributions

The paper makes an empirical contribution rather than proposing a new learning algorithm. It assembles a repro-

ducible pipeline for multi-source Indian housing data, combines coordinate-derived features with a leakage-aware hierarchical encoding, and compares the resulting model across multiple geographic evaluation settings. The experiments also document limitations: locality coordinates are approximate centroids, source distributions differ, and transfer to an entirely unseen city is harder than random-split prediction.

II. Related Work

Hedonic modelling explains observed prices through the implicit values of differentiated property attributes [1]. Spatial statistics further emphasise that observations may exhibit geographic dependence, motivating explicit treatment of spatial structure rather than assuming independent samples [2], [3]. These ideas are directly relevant to housing data, where nearby listings may share market conditions.

Tree ensembles are widely used for tabular prediction. Random forests aggregate decorrelated decision trees [6]; gradient boosting builds additive models by sequentially optimising a loss function [7]. XGBoost improves scalability and regularisation for boosted trees [8], LightGBM introduces efficient histogram-based learning and leaf-wise growth [9], and CatBoost addresses categorical variables through ordered boosting and specialised categorical processing [10]. Ridge regression remains a useful regularised linear baseline [5].

Encoding high-cardinality categorical variables with target statistics can improve predictive performance but may overfit when categories have few observations. Micci-Barreca describes a preprocessing scheme for high-cardinality categorical attributes [11]. This work adapts the general shrinkage idea to a geographic hierarchy: city, spatial cluster, and locality. The encoding is computed out-of-fold for training records so that a row's target does not contribute to its own feature.

Spatial validation is essential when data have temporal, spatial, hierarchical, or phylogenetic structure. Roberts *et al.* discuss cross-validation strategies for dependent data and the risk of over-optimistic estimates when validation does not match the target prediction setting [4]. The study therefore separates random-listing performance from locality- and city-level transfer.

For interpretability, SHAP assigns feature contributions to individual predictions using a unified additive explanation framework [12]. Tree-specific SHAP methods enable efficient attribution for tree ensembles, and later work discusses moving from local explanations to global understanding [13]. In this study, attribution is aggregated into feature families to avoid overinterpreting correlated individual geographic variables.

A. Research gap

Many tabular house-price pipelines compare algorithms using random splits and raw location categories. This leaves three practical questions less visible: (i) whether coordinate-derived features improve prediction beyond property attributes; (ii) whether target-encoded location features remain useful for localities absent from training; and (iii) how much performance degrades when a whole city is unseen. This project addresses these questions through controlled feature ablation and distinct geographic evaluation protocols. The results should be interpreted as evidence for this dataset and pipeline, not as proof that one feature design will generalise to every real-estate market.

III. Dataset and Preprocessing

Four public data sources were used: Bengaluru House Price Data, Hyderabad House Price data, Housing Prices in Metropolitan Areas of India (six city files), and Real Estate Data from Seven Indian Cities. The notebook explicitly maps each source's columns to a common schema instead of relying on fuzzy column-name inference. Price and area units are harmonised before modelling; prices are expressed in Indian lakh rupees (INR lakh) for reporting.

The notebook reports 64,471 standardised rows before cleaning and 55,912 records after cleaning and duplicate/impossible-value filtering. The retained data cover Bengaluru, Chennai, Delhi, Hyderabad, Kolkata, Mumbai, and Pune, with 3,938 unique locality names. The notebook's source-by-city distribution is uneven, which may introduce source and sampling effects. Such differences are considered in source-level diagnostics and are included as a limitation.

A. Cleaning

The preprocessing pipeline normalises city and locality strings, parses numeric property fields, harmonises area and price units, removes impossible or unusable values, and removes duplicates. Model inputs include property attributes such as area, bedroom count, bathrooms, and balcony-related fields where available. Missingness and inconsistent schemas are handled during dataset harmonisation and feature preparation. All predictions are trained on log price, while reported error metrics are calculated on the original price scale unless explicitly identified as log-space R^2.

B. Geocoding and coordinate quality

Localities are geocoded in three stages: OpenStreetMap Nominatim, Photon as a fallback, and a retry using simplified locality names. Geocoding results are cached for reproducibility and resumability. Candidate coordinates are accepted only when they fall within a city-specific radius, reducing the chance of accepting a same-name locality from a different region. The notebook reports coordinates for approximately 96.9% of listings. Coordinates refer to locality-level centroids or representative points, not exact property addresses.

IV. Methodology

Fig. 1 summarises the end-to-end design. The pipeline consists of schema harmonisation, cleaning, locality geocoding, property and geographic feature construction, leakage-aware encoding, model training, and geographic evaluation.

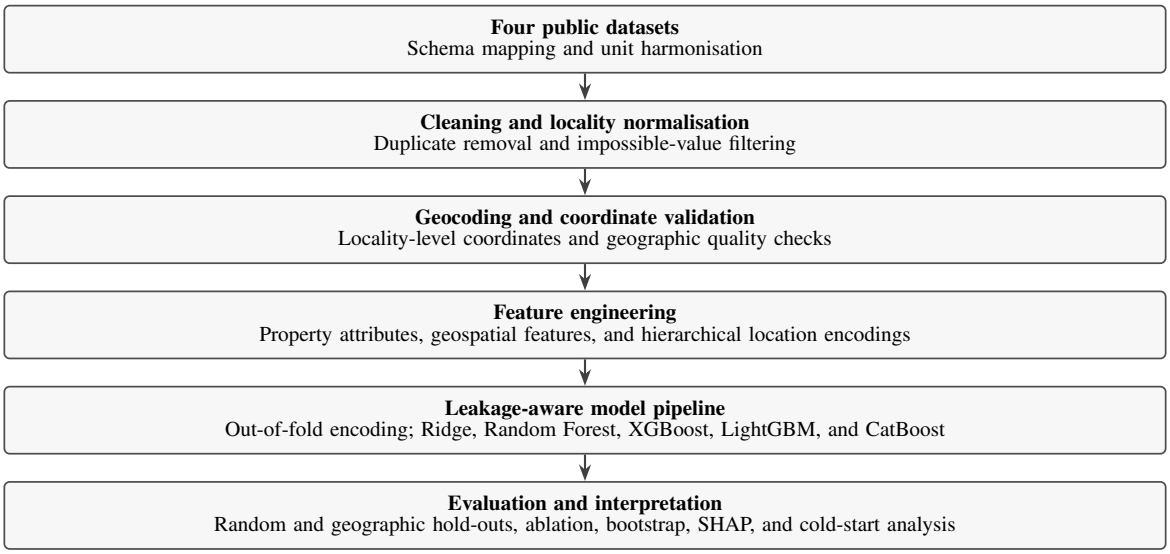


Fig. 1. Overview of the implemented house-price prediction pipeline.

A. Feature groups

The notebook builds several feature families:

- **Property features:** area-derived transformations and available structural attributes such as bedrooms, bathrooms, and balconies.
- **Coordinate-derived features:** distance from the city centre, relative distance normalised by city scale, and coordinate offsets. City-specific absolute offsets are excluded from the leave-one-city-out experiment when they do not transfer to a new city.
- **Spatial density and clusters:** local listing density and the number of nearby localities within a 3 km neighbourhood, plus per-city clusters learned from coordinates without using price targets.
- **Optional amenity features:** proximity or counts for OpenStreetMap amenities such as stations, schools, hospitals, malls, parks, and airports. Availability varies by city and notebook configuration; the main reported results should not be interpreted as demonstrating an amenity effect when these optional features were disabled.
- **Location-price features:** smoothed city, spatial-cluster, and locality price-per-square-foot statistics, plus a distance-weighted price signal from the eight nearest localities.
- **Source provenance:** source indicators are added only in the final ablation feature set to examine dataset-source effects.

B. Hierarchical smoothed location encoding

Let y_i denote the log price per square foot for training listing i. The city-level statistic provides the broad market baseline. The cluster-level statistic is shrunk toward the city estimate, and the locality-level statistic is shrunk toward its cluster estimate. For a child group g with n_g training observations, target sum S_g, parent estimate μ_p, and shrinkage strength m, the encoded value is

$$\mu_g = \frac{S_g + m\mu_p}{n_g + m}. \tag{1}$$

When n_g is large, the local observations dominate; when n_g is small, the estimate moves toward the parent. The shrinkage strengths are tuned using validation performance across random and unseen-locality splits. The notebook's selected values are $m_{\text{loc}} = 80$, $m_{\text{cluster}} = 15$, and $m_{\text{city}} = 5$.

C. Leakage prevention

Target encodings can leak information if a row's own target contributes to the encoded predictor. To reduce this risk, training encodings are computed out-of-fold: each row receives statistics calculated without its fold. Test rows are encoded only using training data. For locality-held-out and city-held-out evaluation, the inner encoding process follows the corresponding geographic split logic so that training features better mimic the information available at inference time. A spatial-neighbour lag is computed from other localities and excludes the locality itself.

D. Learning algorithms

Five regressors are compared: Ridge Regression, Random Forest, XGBoost, LightGBM, and CatBoost. Ridge provides a linear regularised baseline, while the four tree-based methods model non-linear effects and interactions. Models are trained to predict log price, which reduces scale skew. Hyperparameter tuning beyond the documented shrinkage search is not presented as exhaustive; the benchmark reflects the notebook's implemented settings and environment.

E. Feature-set ablation

The LightGBM ablation uses these nested feature sets: A) property-only; B) property plus naive locality encoding; C)

property plus geospatial features; D) property plus hierarchical smoothed encoding; E) geospatial plus hierarchical encoding; F) full proposed features including the spatial-neighbour lag; and G) set F plus source indicators. This design separates the value of raw locality price averages from features that can transfer to unseen localities. In locality-held-out evaluation, naive locality encoding necessarily falls back to the property-only baseline because the held-out locality has no training average.

F. Evaluation protocols and metrics

Three protocols are used:

1) **Random hold-out / random five-fold:** listings are randomly partitioned; this is a conventional benchmark but may share localities across partitions.
2) **Locality-held-out:** entire localities are withheld from training, testing transfer to unseen neighbourhoods.
3) **Leave-one-city-out:** one city is withheld as the test city. Only transferable feature sets A and C are reported because city- and locality-specific price encodings are unavailable for a wholly unseen city.

The principal metric is median absolute percentage error:

$$\mathrm{MdAPE} = 100 \cdot \mathrm{median}_i \left| \frac{y_i - \hat{y}_i}{y_i} \right|, \tag{2}$$

where y_i and $\hat{y}_i$ are observed and predicted prices on the original scale. MAE, RMSE, and R^2 are also reported in the notebook. Lower MdAPE, MAE, and RMSE are better; higher R^2 is better. To quantify uncertainty, the notebook resamples whole localities in a cluster bootstrap rather than treating every listing as independent.

G. Explainability and cold-start analysis

SHAP TreeExplainer is applied to the fitted CatBoost model. Absolute SHAP values are aggregated into property, hierarchical location, geospatial, spatial-neighbour lag, and city-identity families. These scores describe how the fitted model allocates predictive attribution; they are not causal effects. In the cold-start experiment, half of each target city's listings are held out as a fixed test set while the model receives $k \in \{0, 25, 100, 400\}$ labelled listings from that city. At $k = 0$, features that require target-city price statistics are unavailable; with additional examples, the hierarchical encoder can adapt to the new market.

V. Results

A. Model benchmark

Table I and Fig. 2 show the random 80/20 benchmark. The full location-aware feature set reduces MdAPE for every compared model. LightGBM achieves the lowest reported MdAPE for the full feature set (21.77%), while CatBoost has the highest reported log-space R^2 among the full-feature models in the notebook output (0.646).

TABLE I
Random 80/20 hold-out benchmark (values reported by the notebook).

Model	MdAPE A	MdAPE F	$R^2_{\log}$ A	$R^2_{\log}$ F
Ridge	32.8	27.0	0.483	0.573
Random Forest	27.9	22.4	0.532	0.627
XGBoost	27.2	22.1	0.561	0.643
LightGBM	27.2	21.8	0.562	0.642
CatBoost	27.3	22.2	0.570	0.646

A: property-only; F: full proposed feature set including spatial-neighbour lag. MdAPE is in percent. Values rounded to one decimal for readability.

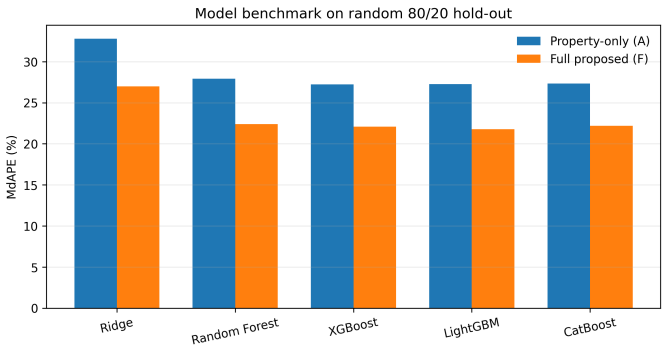


Fig. 2. Model benchmark for property-only and full location-aware features. Lower MdAPE indicates lower typical percentage error.

B. Feature ablation

The LightGBM ablation in Table II and Fig. 3 shows that adding naive locality encoding yields a substantial improvement under random evaluation. Geospatial features alone reduce MdAPE from 27.149% to 21.961%. Hierarchical encoding alone is less effective than geospatial features in this random-split result, while combining both gives 22.056%. The full set reaches 21.955%, a small difference from set C; therefore, the random-split result does not support a claim that every added feature group produces a meaningful additional gain.

C. Geographic generalisation

The random, locality-held-out, and leave-one-city-out results are not interchangeable because they answer different questions. In the notebook, the property-only MdAPE is 27.149% for random evaluation, 28.043% for locality-held-out evaluation, and 38.747% for leave-one-city-out evaluation. Geospatial features reduce the corresponding errors to 21.961%, 26.491%, and 38.543%. Thus, the benefit is clearest for random and unseen-locality evaluation; the improvement for a wholly unseen city is small (0.204 percentage points) and the overall error remains high. The full proposed feature set is not reported for leave-one-city-out because its city/locality target statistics are not defined for an unseen city.

TABLE II
LIGHTGBM FEATURE ABLATION UNDER RANDOM FIVE-FOLD EVALUATION.

Feature set	MdAPE (%)
A: Property only	27.149
B: + Naive locality encoding	22.750
C: + Geospatial features	21.961
D: + Hierarchical smoothed encoding	23.021
E: + Geospatial + hierarchical	22.056
F: Full proposed (+ spatial lag)	21.955
G: F + source indicators	22.038

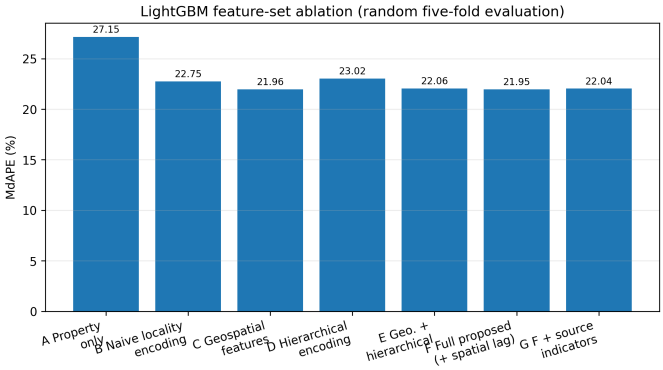


Fig. 3. Ablation of feature groups using LightGBM. The largest improvement over property-only features comes from adding location information; the final small differences should be interpreted with uncertainty estimates.

D. SHAP feature-family attribution

Fig. 5 summarises grouped mean absolute SHAP attribution for the fitted CatBoost model. Property features account for 54.3%, hierarchical location encoding for 30.0%, geospatial features for 11.0%, spatial-neighbour lag for 2.9%, and city identity for 1.7%. The three location-related groups other than city identity account for 43.9% in total. This supports the interpretation that the fitted model uses location signals alongside structural property attributes. Attribution is not causal evidence, and correlated geographic variables may share credit.

E. Cold-start behaviour

The notebook's few-shot analysis evaluates the mean MdAPE as the number of labelled examples in the target city grows. Across the aggregated curve, property-only MdAPE declines from 42.58% at $k = 0$ to 31.89% at $k = 400$. Geospatial features decline from 42.48% to 28.57%, and the full proposed feature set reaches 28.00% at $k = 400$. The full feature set is unavailable at $k = 0$ because its target-dependent encoding cannot be computed for a city with no labelled listings. These findings suggest that a small amount of local labelled data can improve adaptation, although the city-specific curves vary and the aggregated values should not be treated as a guarantee for every city.

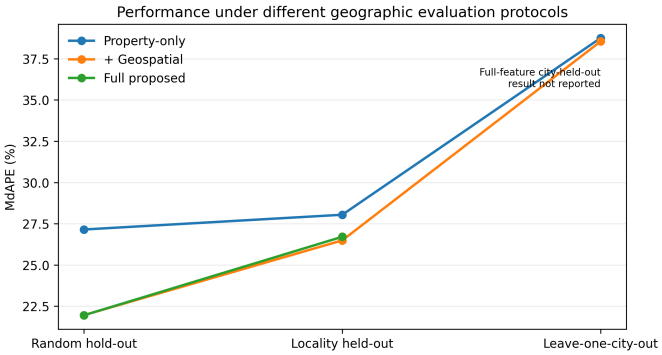


Fig. 4. MdAPE across geographic evaluation protocols. The city-held-out protocol is more difficult and uses only transferable feature sets.

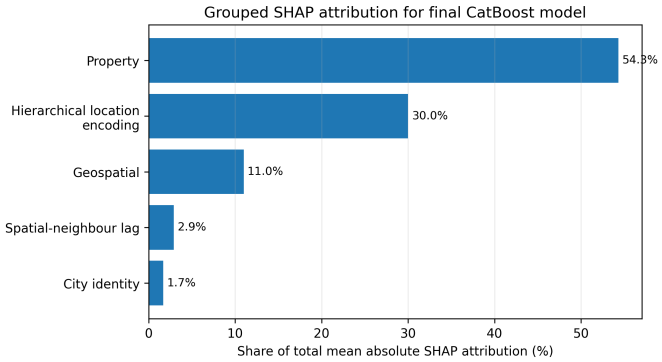


Fig. 5. Grouped mean absolute SHAP attribution for the final CatBoost model. Attribution describes model behaviour, not causal influence on market prices.

F. Uncertainty and diagnostics

The notebook implements locality-cluster bootstrap intervals for random and unseen-locality evaluation. Its random-split output reports the full feature set at MdAPE 21.955% with a 95% interval of [21.11, 22.86]%, compared with property-only at 27.15% [26.07, 28.30]%. The interval for the change relative to property-only excludes zero, supporting an improvement under this protocol. Nevertheless, the comparison between the full set and geospatial-only set is small, so it should not be overstated. Per-city diagnostics also show uneven performance: the notebook reports higher MdAPE for Kolkata (37.18%) and Mumbai (36.35%) than for Hyderabad (14.88%) on the hold-out diagnostics. Such variation motivates reporting subgroup results rather than only a pooled metric.

VI. DISCUSSION

The experiments support three main observations. First, location-aware features improve random hold-out accuracy across all five algorithms, indicating that property structure alone does not capture the geographic variation in the pooled data. Second, coordinate-derived features transfer more naturally to unseen localities than raw locality averages. Third, the leave-one-city-out results expose a clear generalisation challenge: a model trained on six cities does not automatically learn the price level and market structure of a seventh city. The

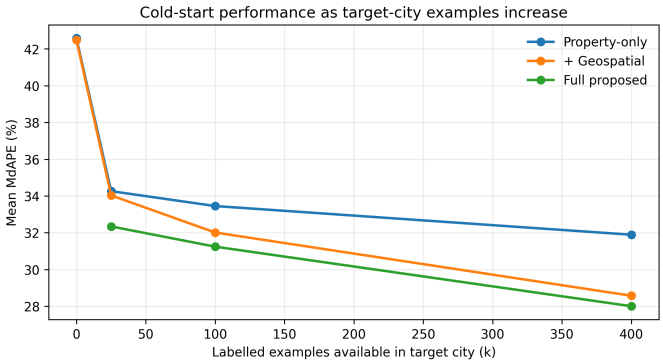


Fig. 6. Cold-start performance as labelled examples become available in a target city. Lower mean MdAPE is better; the full target-encoded feature set is unavailable at $k = 0$.

cold-start experiment suggests that even a modest number of local labelled listings can help adapt the hierarchical encoding.

The difference between the full feature set and the geospatial-only set under random evaluation is small. The appropriate claim is therefore not that every engineered feature independently improves accuracy, but that location features collectively provide substantial benefit over the property-only baseline, while the incremental benefit of the spatial-neighbour lag and source indicators is limited in the reported random-split ablation. This distinction is important for reproducibility and for avoiding overstated novelty claims.

VII. Limitations and Threats to Validity

Several limitations must be considered. First, the source datasets differ in coverage, collection time, and listing conventions. Source indicators help diagnose these differences but cannot fully remove distribution shift. Second, locality centroids are approximate and may not represent the exact location of each property. Third, listing density is a proxy based on observed listings and should not be interpreted as complete housing supply. Fourth, OpenStreetMap amenity data may be incomplete or uneven across cities; the notebook notes that amenity features can be disabled, so claims about their impact require confirming the configuration used for the final run. Fifth, the two location shrinkage parameters are tuned on the same overall dataset later used for evaluation, making the final estimates slightly optimistic. Sixth, random-split evaluation can be optimistic when listings from the same locality appear in training and testing. Finally, the leave-one-city-out experiment can evaluate only features that remain meaningful for an unseen city; city-specific target encodings are intentionally excluded.

Before submission, the team should rerun the notebook from a clean runtime, confirm that every result table corresponds to the same final configuration, and verify all cited publications against their original sources.

VIII. Conclusion and Future Work

This paper examined location-based feature engineering for house-price prediction across seven Indian cities. From 55,912 cleaned records, the implemented pipeline combined property attributes, coordinate-derived descriptors, hierarchical smoothed location-price statistics, and spatial-neighbour information. In the random 80/20 benchmark, LightGBM's MdAPE decreased from approximately 27.15% with property-only features to 21.77% with the full feature set. Ablation results indicate that geospatial features provide a strong and transferable signal, while locality-held-out and leave-one-city-out results show that geographic generalisation remains a harder problem than random-split prediction. SHAP attribution indicates that location-related feature families contribute materially to the fitted model, though the attribution should not be interpreted causally.

Future work should use a strictly nested geographic validation design for tuning and final testing, add time-aware evaluation when listing dates are available, improve coordinate precision, verify amenity completeness, and evaluate transfer to cities entirely absent from model development. A future release should also provide dataset provenance, a fixed environment file, and the exported result tables so that reported numbers can be independently reproduced.

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